

UCE822 :URBAN TRANSPORTATION PLANNING				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To expose the students to various concepts of transportation planning on urban platform, Land use planning, use of transportation modelling in travel demand management and public transit studies				
The Transportation Planning Process and Problems: Terminology of Transportation Planning, Functional Components, Brief Overview of Models used in Transportation Planning, Stakeholder input in the planning process, Environmental concerns, Smart growth and sustainable alternatives, Energy based planning, Intelligent Transportation Systems, Global Positioning Systems.				
Transportation System Impacts: Travel Facilities, Origin and Destination, Transit Surveys Decision making Process, Transportation Demand Management (TDM).				
Modeling: Transportation system characteristics and interrelationships, User costs and Human Activities, Travel Demand Forecasting, Trip Generation, Trip Distribution, Modal Choice, Trip Assignment.				
Land Use Transportation System: Urban system components, Urban Spatial Structure, Location Theory, Land use planning, Land use Models, Land use transport models – (Lowry and Garin), Lowry Models, Transit Oriented Development(TOD).				
Urban Public Transportation: Urban Growth and Public Transport needs, Transit mode characteristics, transit characteristics, Fleet size and capacity estimation, Smart cities based Transit Planning				
Project assignment/ Micro project: 1. Study area delineation 2. Travel Survey Design 3. Home interview survey 4. Data analysis 5. Use of Transportation Simulation Software like VISUM/ TransCAD. 6. Case study: Planning for transportation systems of a specific location				
Course Learning Objectives (CLO) Upon completion of this course, the students will be able to: 1. Understand the concept of Transportation Planning in the Regional and City level Planning 2. Design Travel Demand and Transit based Surveys. 3. Estimate Travel Demand of a particular corridor, city or area 4. Perform Four Stage Modelling for Travel demand and further Design transportation alternatives for the same. 5. Suggest mass transit alternatives for a given conditions on the base of its characteristics and capacity estimate				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25